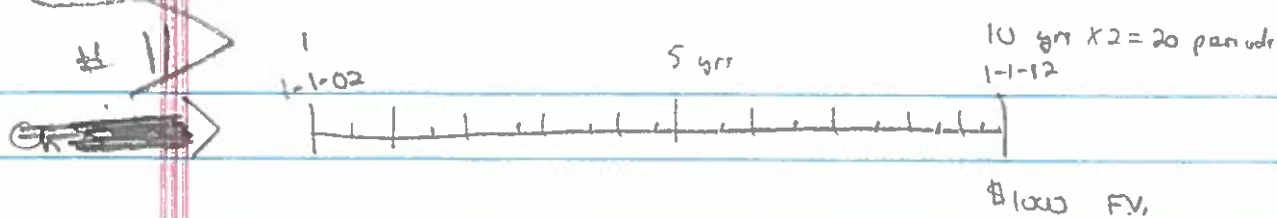


BONDS PROBS 527

BONDS PROBS 527



$$PMT = 4\% \div 2 \times \$1000 = \$20$$

$$N = 10 \text{ yrs} \times 2 = 20 \text{ periods}$$

$$i = 12\% \div 2 = 6\% \text{ semi}$$

$$FV = \$1000$$

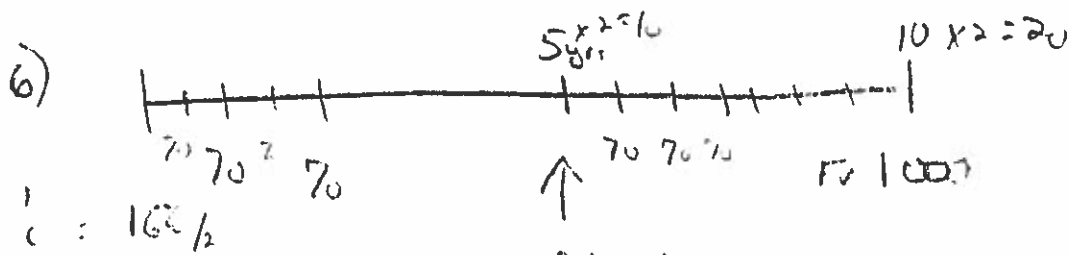
$$PV = ? = \$541.20$$

$$\text{Book Value of Bonds} = \$10,000,000$$

$$\text{Bond Par Value} = \$1000$$

$$\begin{aligned} &= 10,000 \text{ Bonds} \\ &\times \text{Market price} \times \$541.20 \\ &\underline{\underline{\$5,412,000}} \end{aligned}$$

CH 5 BOND P.S.F.



$$PMT = 70$$

$$i = 16\% / 2 = 8\% \text{ currently}$$

$$FV = 1000 \rightarrow 1073.60$$

$$N = 20 = 10$$

$$PV = ?$$

$$\text{Today} = \$966.99$$

$$PV = ?$$

$$FV = 1000$$

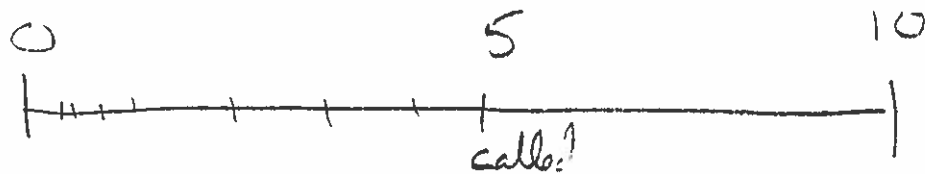
$$N = 5 \times 2 = 10$$

$$i = 12\% / 2 = 6\%$$

$$PMT = 70$$

$$PV = ? = 1073.60$$

7) CH 5 BOND P-58



C/PN Rate : 10%, compounded
 $= 2.5\% (\$1000) 8.1\%$

FV = \$1000 =

PMT: 25

YTM: 8%

YTC: 7.5%

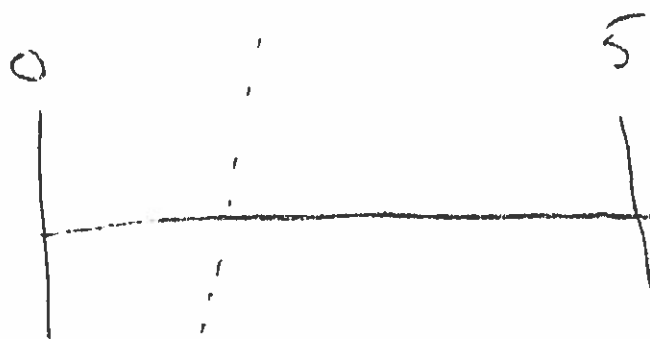
FV = \$1000

PMT = 25

N = 10 x 4 = 40

i = 8% / 4 = 2

PV = ? = 1136.78



PV = 1136.78

N = 5 x 4 = 20

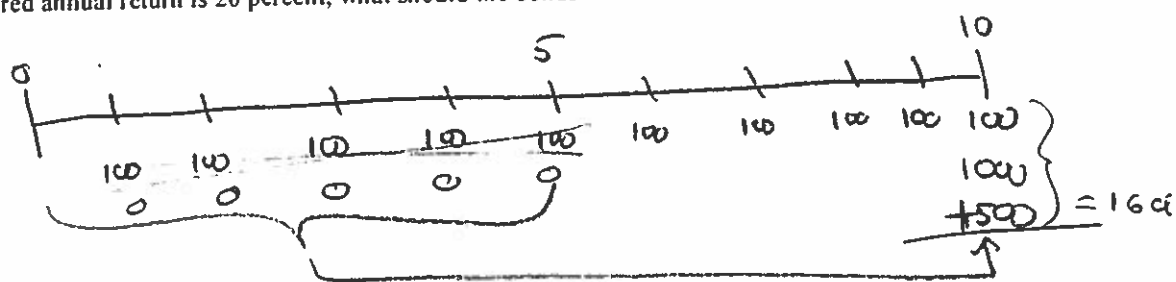
PMT = 25

i = 7.5 / 4 = 1.875%

FV = ? = 1048.34

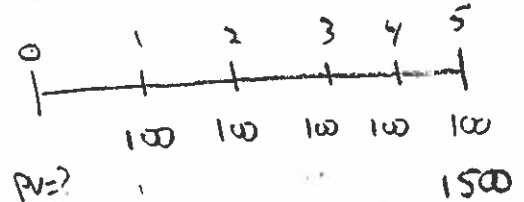
Call Price

Recently, Ohio Hospitals Inc. filed for bankruptcy. The firm was reorganized as American Hospitals Inc., and the court permitted a new indenture on an outstanding bond issue to be put into effect. The issue has 10 years to maturity and a coupon rate of 10 percent, paid annually. The new agreement allows the firm to pay no interest for 5 years. Then, interest payments will be resumed for the next 5 years. Finally, at maturity (Year 10), the principal plus the interest that was not paid during the first 5 years will be paid. However, no interest will be paid on the deferred interest. If the required annual return is 20 percent, what should the bonds sell for in the market today?



$$\begin{aligned}
 CF_0 &= 0 \\
 CF_1 &= 0 \quad 5 \text{ N} \\
 CF_2 &= 100 \quad 4 \text{ N} \\
 CF_3 &= 1600 \\
 I &= 20 \\
 NPV &= ? = \underline{362.44}
 \end{aligned}$$

$$\begin{aligned}
 CF_0 &= 0 \\
 CF_{1-5} &= 0 \\
 CF_{6-9} &= 100 \\
 CF_{10} &= 1600 \\
 i &= 20\% \\
 NPV &= ? = \underline{362.44}
 \end{aligned}$$



$$\begin{aligned}
 PV &= ? \\
 PMT &= 100 \\
 N &= 5 \\
 i &= 20 \\
 FV &= 1500 \\
 PV &= ? = 901.88 \text{ (B)}
 \end{aligned}$$



$$PV = 362.44$$

$$\begin{aligned}
 FV &= 901.80 \\
 N &= 5 \\
 i &= 20 \\
 PV &= ? = \text{(3)}
 \end{aligned}$$

Q1. Zero Coupon Bonds

Suppose your company needs to raise \$10 million and you want to issue 20-year bonds for this purpose. Assume the required return on your bond issue will be 9 percent, and you're evaluating two issue alternatives: a 9 percent annual coupon bond, and a zero coupon bond. Your company's tax rate is 35 percent.

- a. How many of the coupon bonds would you need to issue to raise the \$10 million? How many of the zeroes would you need to issue?
- b. In 20 years, what will your company's repayment be if you issue the coupon bonds? What if you issue the zeroes?
- c. Consider the firm's after tax cash flow for the first year under the two scenarios. Why would you want to ever issue zero coupon bonds?

4.6 Req'd Return = 9% YTM = 9% Mkt int. rate

$\Downarrow$ =

9% coupon rate so trades at Par \$1000.

$$\frac{\text{Need } \$10,000,000}{\text{Price } \$1000} = 10,000 \text{ bonds}$$

Zeros

$$FV = 1000$$

$$N = 20$$

$$i = 9$$

$$PMT = 0$$

$$PV = ? = \$178.43$$

$$\frac{\$10,000,000}{\$178.43} = 56,044 \text{ bonds}$$

b) coupon repayment \$ amt @ maturity in 20 yrs



\$10,000,000 FV

9% PMT & int

\$10,900,000 Repaid

$$[(.09)(10,000,000)]$$

Zeros

$$\begin{array}{r} 56,044 \text{ bonds} \\ \times \$1000 \\ \hline \$56,044,000 \text{ repaid} \end{array}$$

c) Coupon Bonds

10,000 bonds
 $\times \$ 90$ int pd 1st yr
 $\$ 900,000$ per year interest exp
 $(1 - .35)$ After tax effed
 $\$ 585,000$ AT cash outflow

zeroes



$R/N = 20$

$Pr = 178.43$

$\$ 16.06$ interest $\times$ ^{BONDS} $(56,044) (.35) = 315,023$ cash inflow

1775

BOND

PROB 577

#12

~~Ch 8-101~~Burger King annual

$$\text{Par} = \text{FV} = \$1000$$

$$\text{Coupon} = 8\% \times \$1000 = \$80 = \text{PMT}$$

$$N = 20$$

$$i = ? =$$

McDonald's

semi-annual

$$\text{Par} = \text{FV} = \$1000$$

$$\text{Coupon} = 8\% / 2 = 4\% \times \$1000 = \$40 = \text{PMT}$$

$$N = 20 \times 2 = 40$$

$$i = 12\% / 2 = 6\%$$

$$\text{PV} = ? = \$699.07$$

Use Equiv. Annual Rate based on

Mickey D's 12% nominal-req'd

return on its semiannual basis bonds

$$\text{So } 12 = \text{NOM}$$

$$2 = \text{P/Yr}$$

$$? = \text{EFF} = \text{EAR} = 12.36\%$$

So

$$\text{FV} = \$1000$$

$$\text{PMT} = \$80$$

$$N = 20$$

$$i = 12.36$$

$$\text{PV} = ? = \$681.54$$

$$\text{McD's } 699.07$$

$$\text{B.K.} - 681.54$$

$$= \$17.53$$

$$ROE_p = 6\%$$

$$DEBT = \$200,000$$

$$DEBT\ RATIO = \frac{D}{A} = 80\% = \frac{200,000}{Assets} \quad \text{so } Assets = \frac{200,000}{.8} \text{ or } \$250,000$$

$$L_N\% = 10\%$$

$$i_o\% = 14\%$$

$$Sales_p = \$300,000$$

$$EBIT = 11\% (Sales) \Rightarrow .11 (\$300,000) = \$33,000 = EBIT$$

$$Tax = 40\%$$

$$ROE_N = ?$$

$$AND \quad A = D + E$$

$$\$250,000 = \$200,000 + \boxed{?} \$50,000$$

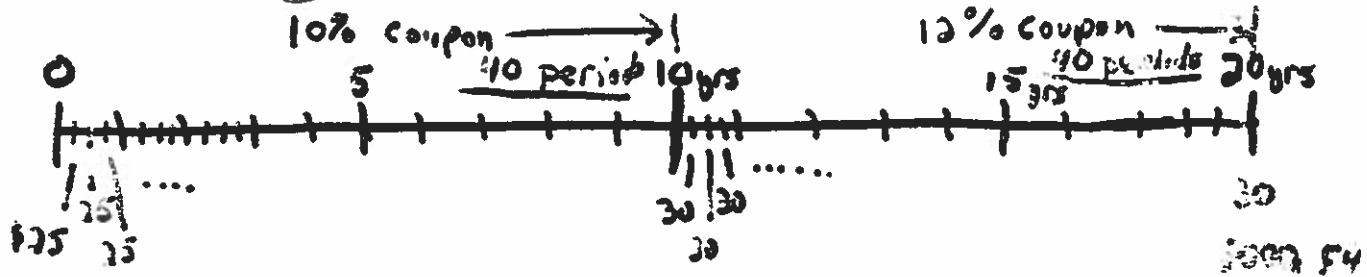
EBIT _N =	\$33,000	
- Int. Exp	- 20,000	
	13,000	
= EBT		
- Tax 40%	- 5,200	
	7,800	
= NI		

$(i_o\%)(D)$
 $(.10)(200,000)$

$$ROE_N = \frac{NI}{Eq_y} = \frac{7,800}{50,000} = 15.6\%$$

10. Bond vs. Mortgage

Suppose Hadden Inc. is negotiating with an insurance company to sell it a bond issue. Each bond has a par value of \$1,000, it would pay 10% per year in quarterly payments of \$25 per quarter for 10 years, and then it would pay 12% per year (\$30 per quarter) for the next 10 years (Years 11-20). The \$1,000 principal would be returned at the end of 20 years. The insurance company's alternative investment is in a 20-year mortgage which has a nominal rate of 14 percent and which provides monthly payments. If the mortgage and the bond issue are equally risky, how much should the insurance company be willing to pay Hadden for each bond?



Need i
to discount
Bond $\Rightarrow PV = ?$

MORT **BOND**
 $\text{Eff} = 14.93 \Rightarrow = ? \text{ quarterly nominal rate?}$

Know $\text{Eff} = 14.93$ **MORT** **BOND**

for Bonds $P/Yr = 4$

$\text{Nom?} = ? = 14.16\%$ **BOND**

If this is nominal rate set on bond issue,
 then insurance co. will earn same
 effective rate as it can get on mortgage.

WANT PERIODIC RATE ON BOND TO DELT
BOND C.F.'S

Periodic rate for 14.16% nom. rate with
 quarterly compounding?

$$\frac{14.16}{4} = 3.54\% \text{ per } \begin{matrix} \text{qtrly} \\ \text{period} \\ \text{on} \\ \text{bonds} \end{matrix}$$

So: $CF_0 = 0$

$CF_1 = 25$; #Times = $40 = N$

$CF_2 = 30$; " " = $39 = N$

$CF_3 = 1030$ (FV & last int pmt)

$I = 3.54$

$\text{NPV} = ? = \$750.78$

so invest in bond
 if it can be BPT
 for $\leq \$750.78$

BONDS

Par V. \$1000

Stated Rate = $10\% \div 4 =$
Qrtly $(2.5\% \times \$1000)$

PMTs = \$25

for 10 yrs or 40 periods

Then stated = $12\% \div 4 =$
Qrtly $3\% \times \$1000$

PMT = \$30

for yrs 11-20
or periods 41-80

MORTGAGE

Nom i = 14%
monthly

Nom rate of i $\Rightarrow$ eff

NOM = 14%

P/yr = 12

EFF = ? = 14.93%
on Mortg.

FIND EQUIVALENT RATE ON BONDS

MORTG

14.93%

Equivalent Bond
EFF% $\Rightarrow$ 14.93%

P/yr = 4

NOM = ? = 12.16% on BONDS

Qrtly periodic = $\frac{12.16}{4} = 3.04\%$

USE to
Dset Bond
CFs

$$CF_0 = 0$$

$$CF_1 = 25$$

$$\# \text{ TIMES} = 40 = N$$

$$CF_2 = 30$$

$$\# \text{ TIMES} = 39 = N$$

$$CF_3 = 1030$$

$$i = 3.54\%$$

$$NPV = ? = \$ 750.78$$

So invest in
bond if can buy
for $\leq \$ 750.78$

5. A 10-year bond has a 10 percent annual coupon and a yield to maturity of 12 percent. The bond can be called in 5 years at a call price of \$1,050 and the bond's face value is \$1,000. Which of the following statements is most correct?

a. The bond's current yield is greater than 10 percent.
b. The bond's yield to call is less than 12 percent.
c. The bond is selling at a price below par.
☒ d. Both answers a and c are correct.
e. None of the above answers is correct.

6. Assume that you are considering the purchase of a \$1,000 par value bond that pays interest of \$70. each six months and has 10 years to go before it matures. If you buy this bond, you expect to hold it for 5 years and then to sell it in the market. You (and other investors) currently require a nominal annual rate of 16 percent, but you expect the market to require a nominal rate of only 12 percent when you sell the bond due to a general decline in interest rates. How much should you be willing to pay for this bond?

a. \$842.00
b. \$1,115.81
c. \$1,359.26
☒ d. \$966.99
e. \$731.85

7. Kennedy Gas Works has bonds which mature in 10 years, and have a face value of \$1,000. The bonds have a 10 percent quarterly coupon (i.e., the nominal coupon rate is 10 percent). The bonds may be called in five years. The bonds have a nominal yield to maturity of 8 percent and a yield to call of 7.5 percent. What is the call price on the bonds?

a. \$ 379.27
b. \$1,025.00
☒ c. \$1,048.34
d. \$1,036.77
e. \$1,136.78

8. Bankruptcy Re-Org and Debt Re-Structuring

Recently, Ohio Hospitals, Inc. filed for bankruptcy. The firm was reorganized as American Hospitals, Inc., and the court permitted a new indenture on an outstanding bond issue to be put into effect. The issue has 10 years to maturity and a coupon rate of 10 percent, paid annually. The new agreement allows the firm to pay no interest for 5 years. Then, interest payments will be resumed for the next 5 years. Finally, at maturity (Year 10), the principal plus the interest that was not paid during the first 5 years will be paid. However, no interest will be paid on the deferred interest. If the required annual return is 20 percent, what should the bonds sell for in the market today?

9. Zero Coupon Bonds

Suppose your company needs to raise \$10 million and you want to issue 20-year bonds for this purpose. Assume the required return on your bond issue will be 9 percent, and you're evaluating two issue alternatives: a 9 percent annual coupon bond, and a zero coupon bond. Your company's tax rate is 35 percent.

- How many of the coupon bonds would you need to issue to raise the \$10 million? How many of the zeroes would you need to issue?
- In 20 years, what will your company's repayment be if you issue the coupon bonds? What if you issue the zeroes?
- Consider the firm's after tax cash flow for the first year under the two scenarios. Why would you want to ever issue zero coupon bonds?

10. Bond vs. Mortgage

Suppose Hadden Inc. is negotiating with an insurance company to sell it a bond issue. Each bond has a par value of \$1,000, it would pay 10% per year in quarterly payments of \$25 per quarter for 10 years, and then it would pay 12% per year (\$30 per quarter) for the next 10 years (Years 11-20). The \$1,000 principal would be returned at the end of 20 years. The insurance company's alternative investment is in a 20-year mortgage which has a nominal rate of 14 percent and which provides monthly payments. If the mortgage and the bond issue are equally risky, how much should the insurance company be willing to pay Hadden for each bond?

11. MV of Debt

In order to accurately assess the capital structure of a firm, it is necessary to convert its balance sheet figures to a market value basis. KJM Corporation's balance sheet as of today, January 1, 2009 is as follows:

20 - 1M 4% 500000	Long-term Debt (bonds, at par)	\$10,000,000
1000 = FV	Preferred stock	2,000,000
30 = 100 - 70	Common Stock (\$10 par)	10,000,000
30 = 100 - 70	Retained Earnings	4,000,000
30 = 100 - 70	Total Debt & Equity	\$26,000,000

The bonds have a 4 percent coupon rate, payable semiannually, and a par value of \$1,000. They mature on January 1, 2019. The yield to maturity is 12%. What is the current market value of the firm's debt?

12. Comparing Bond Prices

Assume McDonald's and Burger King have similar \$1,000 par value bond issues outstanding. The bonds are equally risky. The Burger King bond has an annual coupon rate of 8% and matures 20 years from today. The McDonald's bond has a coupon rate of 8 percent with interest paid semiannually, and it also matures in 20 years. If McDonald's has a nominal required rate of return of 12% on its semiannual basis bonds, and Burger King has the equivalent annual rate on its bonds, what is the difference in current market prices of the two bonds?